

Feasibility study of methanol production plant from hydrogen and captured carbon dioxide



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ABSTRACT

This paper aims to present a feasibility study of the innovative plant for methanol synthesis from carbon dioxide-sequestered by fossil fuel power plant and hydrogen, which is produced by water electrolyzer employing the over-production on the electrical grid. The thermo-economic analysis is performed in the framework of the MefCO₂ H2020 EU project and it is referred to the German economic scenario, properly taking into account the real market costs and cost functions for different components of the plant. Three different plant capacities for methanol production (4000 10,000 and 50,000 ton/year) have been investigated, assuming an average cost for electrical energy to feed electrolyzers and analyzing the influence of the most significant parameters (oxygen selling option, methanol selling price and electrolyzers' capital cost) on the profitability of the plant.

The analysis has been performed in W-ECOMP, software for the thermo-economic analysis and plant optimization developed by the University of Genoa.

1. Introduction

Greenhouse gas (GHG) emissions are one of the most important environmental issues of the twenty first century. The largest source of GHG is the carbon dioxide and its emission has tremendously increased in the last decades, mainly due to fossil fuels combustion for the power generation and the automotive transportation.

Moreover, the global energy demand is expected to double by the 2050 and the fossil fuels exploitation will be still predominant in comparison with the renewable energy penetration that, anyway, is increasing as well [1]. In consideration of this, the European Commission is adopting more and more stringent environmental regulations that push the researchers to study innovative systems for the CO₂ emissions reduction by developing new fuels, which have low carbon footprint for the energy production.

Another important aspect is related to the increasing renewable installed capacity in the energy supply grid and its integration with traditional power plants: the strongly stochastic and intermittent nature of the renewable energy sources (RES), in particular wind and solar, makes the power output variable and unpredictable. This translates into issues like system balancing and capacity adequacy that, in turn, result in difficult operating management strategy for traditional power plants and economic losses as well. The strong development of unpredictable RES, such as solar and wind, which has the priority of dispatchment to the energy market, has caused significant issues to traditional power

plants (i.e. combined cycles): forcing them to operate in strong off-design conditions at lower efficiencies and frequent startup/shutdown that reduce their lifetime as well.

The power-to-liquid systems seem to represent a valiant solution for the future energy scenarios: a PtL technology concerns a process that is able to absorb energy (i.e. the overproduction of RES on the grid), converting and storing that energy in chemical form.

The conversion of renewable energy into the more convenient form of liquid energy carriers can be an effective way to moderate the RES intermittency and stabilize the electrical grid, thus avoiding the continuous shut downs that are affecting a number of traditional power plants. Moreover, in comparison with the gas form, the liquid form presents less problems in transportation from the both safety and infrastructure point of view.

One of the most promising products for the PtL systems is methanol (formula CH₃OH): it presents liquid form at atmospheric condition; the melting and boiling point are −97.6 °C and 64.7 °C, respectively.

Methanol is one of the most important building blocks in chemical industry; it is used as feedstock to synthesize chemicals derivatives such as formaldehyde, MTBE and acetic acid, which, in turn, are used in products like adhesives, subfloors, solvents, washer fluid etc...[2].

Moreover methanol is largely used in energy-related applications (about 40% of world production). Methanol presents excellent combustion properties: despite its energy density is about half of the gasoline energy density (20.1 MJ/kg for methanol, 44.3 MJ/kg for

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Nomenclature

Abbreviations

CCS	Carbon capture sequestration
DPBP	Discounted pay back period
GHG	Greenhouse gas
NPV	Net present value

PEC	Purchased equipment cost
PEM	Polymer electrolyte membrane
PtL	Power to liquid
RES	Renewable energy sources
TCI	Total capital investment
TPG	Thermochemical power group
W-ECOMP	Web-based economic cogeneration modular program

gasoline) it has a higher octane number in comparison of gasoline (108 for methanol, 95 for gasoline), that allows higher compression ratio and an increase in combustion efficiency [3,4].

Methanol can be used directly as a vehicle fuel or blended with gasoline to produce a high-octane efficient fuel with lower emissions than conventional gasoline [4]; it is also emerging as a clean-burning marine fuel pushed by the increasingly stringent emissions regulations. Methanol is also a key component in biodiesel, a renewable fuel that can be used in place of conventional diesel fuel, or in blended form [5].

Currently, on the industrial scale, methanol is predominantly produced from natural gas by steam reforming or coal gasification: the resulting syngas is mostly composed by CO, CO₂ and H₂, in correct proportion for the methanol synthesis. The syngas-based methanol production consumed a large amount of fossil resources and inevitably it generates large amounts of carbon dioxide: for each ton of methanol produced in conventional process, about 0.6–1.5 tons of CO₂ are emitted in atmosphere [6,7,8].

As alternative to the traditional approach, an innovative sustainable method for methanol production from RES is investigated in this research paper: methanol is synthesized from a mixture of hydrogen and carbon dioxide. The hydrogen is produced by water electrolysis employing renewable electrical energy or absorbing the overproduction of the grid and CO₂ needed in the reaction can be sequestered from the flue gas of traditional power plant [9,10]. This process represents an eco-friendly solution for the methanol production by allowing reducing the intensity of fossil fuel consumption and mitigating the CO₂ emission as well. The reaction is reported below:



The catalytic reaction takes place in ranges of temperature and pressure of 250–300 °C and 50–100 bar, respectively on CuO/ZnO/Al₂O₃ as catalyzer [11,12].

This paper work has been carried out within the context of the E.U. MeFco2 (methanol fuel from CO₂) project (accepted in SPIRE framework of the Horizon 2020 EU Research and Innovation program) [14]. The MeFco2 aims to develop and design an innovative methanol production technology with a low carbon footprint: the overall project concept is based on the sequestration of carbon dioxide emitted by traditional plant and its use in reaction with hydrogen, which is

produced from surplus electrical energy. Together with the University of Genoa, important academic and industrial partners are involved in the project. The main activities of the research project are (i) development of innovative catalyst for the methanol reaction from H₂ and CO₂; (ii) development of innovative PEM electrolyser for hydrogen production; (iii) design and installation of demonstrative pilot plant and grid integration; (iv) process optimization and thermo-economic analysis of the plant considering different economic scenarios. Unige is involved in the fourth activity [14].

The German economic scenario is chosen as a reference case, since Germany presents the largest installed capacity of unpredictable RES among the European Countries: in 2014 Germany had the record in terms of both wind capacity (40.4 GW, 31% of all the EU28 Countries) and solar capacity (38.4 GW, 43% of all the EU28 Countries) [15]. The impact of RES has become more and more relevant in the last years in particular the production from solar, which was negligible in 2007, increased up to 35 TWh in 2014 and the production from wind has increased from 39.5 to 56 TWh (Fig. 1). On the other hand, the production from thermal and nuclear plants has decreased because of the economic crisis, the electricity demand has slightly decreased (from 636.4 to 612.6 TWh), although the process is less evident compared to other European Countries. It is worth noting that, starting from 2012, the difference between production and demand corresponds to the production of uncontrollable RES (wind and solar).

In this paper, a feasibility study of a power-to-fuel plant for methanol production is performed throughout a thermo-economic analysis. Three different capacity plants for methanol production (4000, 10,000 and 50,000 ton/year) are investigated, assuming an average cost for electrical energy to feed electrolyzers and analyzing the influence of the most significant parameters (oxygen selling option, methanol selling price and electrolyzers capital cost) on the profitability of the plant.

The study is performed using the W-ECOMP software (Web-based Economic Cogeneration Modular Program) for time-dependent thermo-economic optimization, developed by the University of Genoa. W-ECOMP is a tool for economic feasibility evaluation of innovative plant, coupled with the management strategy optimisation. It is based on a genetic algorithm for the minimization of a target function, which is representative of the variable cost of the plant. The first step for the

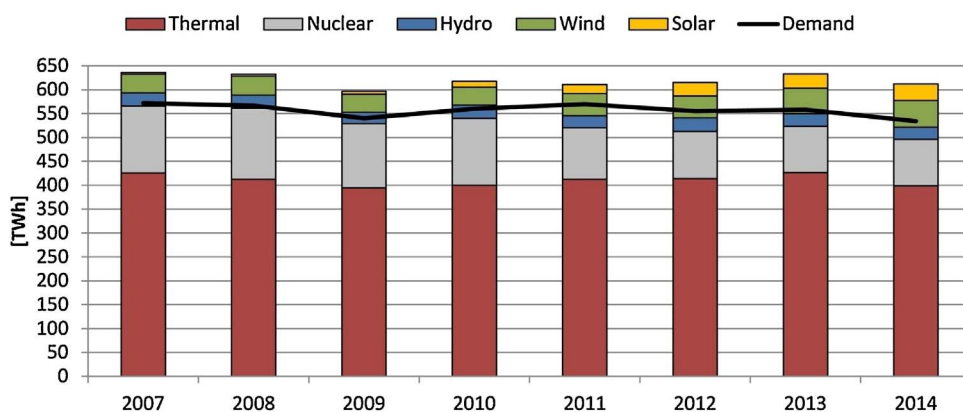


Fig. 1. Energy mix production and electrical demand evolution in Germany (2007–2014) [15].

economic analysis is to determine the Total Capital Investment (TCI): its value depends on the Purchased Equipment Cost (PEC, defined by the cost functions of the modules in the plant under investigation) and other terms that take into account the construction and installation costs (direct costs), the start-up cost and other outlays (working capital, licensing, allowance etc). The final aim is the calculation of the investment's profitability in order to choose the best solution, properly taking into account the initial investment and the associated risks related to the economic scenario and characteristic of the plant investigated. The most important parameters that describe the profitability of the plant are Net Present Value (NPV) and Discounted Pay Back Period (DPBP) calculated as follow:

Net present value:

$$NPV = \sum_{j=1}^{plantlife} \left(\frac{CFN_j}{(1+t)^j} \right) - TCI \quad (2)$$

Discounted payback period

$$TCI = \sum_{j=1}^{DPBP} \left(\frac{CFN_j}{(1+t)^j} \right) \quad (3)$$

The calculation is carried out by dividing the operational time (usually a year) with sufficient number of representative periods (one hour or less depending on the particular application).

More information about the W-ECOMP software can be found in other papers by the authors [16–20].

2. Plant lay-out

The system investigated in this work consists of a plant for the methanol production starting from hydrogen, which is produced by water electrolyser powered by electrical energy from the grid, and carbon dioxide, which is sequestered from a traditional power plant through an amine based CCS plant. A simplified scheme of the plant is reported in Fig. 2.

The “power to methanol” process that is the object of the thermo-economic analysis consists of the following main components:

- (i) Water electrolyser: the device employs electrical energy in the

Table 1

Main technical parameters [10–13,21,22,28].

PEM Electrolyser	
Electrical consumption	5.2 kWh/Nm ³ di H ₂
Pressure	30 bar
Temperature	80 °C
Efficiency	68%
Carbon Capture system	
Treatment kind	Amines (MEA) (30%)
Flue gases inlet T[°C] and p[bar]	40 °C, 2 bar
Thermal energy consumption per ton of CO ₂	3 GJ _{th} /kgCO ₂
CO ₂ outlet temperature[°C] pressure[bar]	40 °C, 2 bar
CO ₂ capture rate	90%
Methanol Reactor	
Working Pressure	80 bar
Temperature	240 °C
Recirculation factor of unreacted syngas	0.85
Conversion efficiency	96%
Molar H ₂ :CO ₂ ratio	3:1
Compressor	
Isentropic efficiency	86%
Mechanical efficiency	99%

process of water electrolysis to produce hydrogen for the methanol synthesis; furthermore a significant amount of oxygen (about 8 kg for each kg of hydrogen produced) is co-produced in the process.

- (ii) Carbon capture system: the amine based CCS is connected to the coal-fired power plant and sequesters the CO₂ required by the methanol production process from the flue gases.
- (iii) Methanol reactor: the mixture of hydrogen and carbon dioxide is sent to the reactor for the methanol production.

Before performing the thermo-economic analysis, the operating parameters of the plant must be defined in order to calculate the mass flows, the electrical consumption and the thermal energy input and output.

The thermodynamic calculations refer to the unitary plant size with 1 MW of installed electrolyser, the results are then scaled up to simulate larger production plants. The thermodynamic assumptions used for the calculation are reported in Table 1: the data are taken from literature or

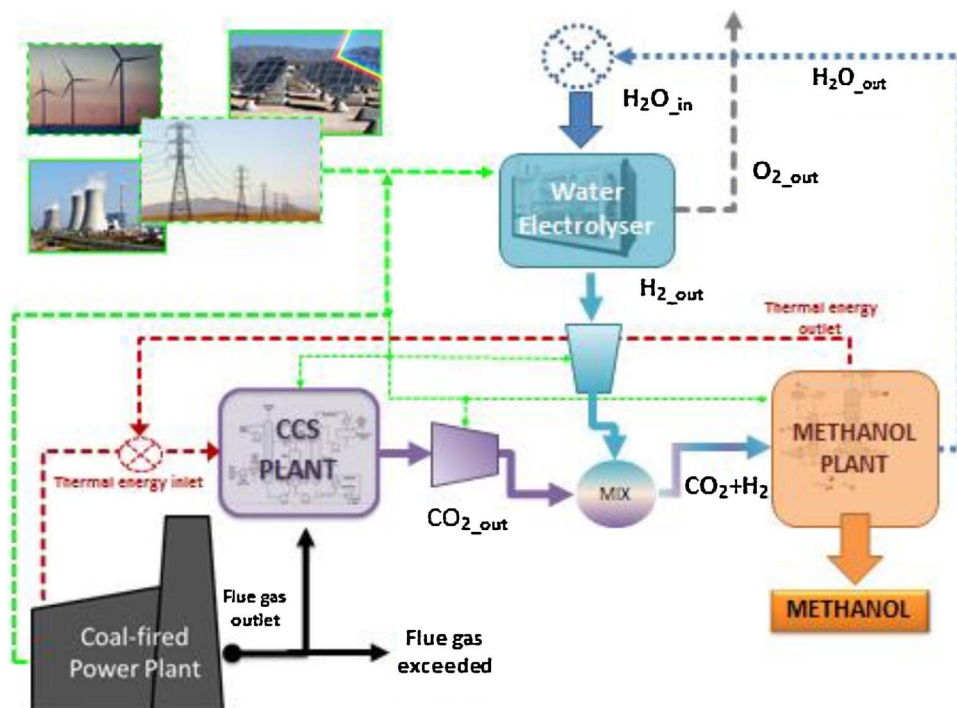


Fig. 2. Simplified plant lay-out.

real commercial information provided by industrial partners of the project.

In reference to the parameters reported in Table 1, 1 MW electrolyser produces about 19 kg/h of hydrogen and 151 kg/h of oxygen (H₂:O₂ ratio is 8). Considering the stoichiometric methanol reaction for 19 kg/h of hydrogen about 140 kg/h of CO₂ are needed: the CCS system is sized in order to be able to capture the required amount of CO₂, hence it is able to process about 824 kg/h of flue gases, assuming the CO₂ average content equal to 17% in mass [21].

The CO₂ exits the CCS at 2 bar, subsequently it is pre-compressed up to 30 bar before being mixed with the hydrogen. Then hydrogen and carbon dioxide are mixed together and compressed to the reactor working pressure, equal to 80 bar.

The results of the thermodynamic analysis in terms of mass flows, electrical energy consumption and thermal energy input and output, are summarized in Table 2.

For the thermo-economic analysis the previous results are implemented in the single modules, representative of each component constituting the plant.

Thermo-economic analysis is carried out on a complete year time period, taking into account the time dependent nature of the problem related to variation of renewable electricity available on the grid. The aim of the study is the thermo-economic optimization of the whole system, the comparison of different plant capacities and the definition of the best plant size taking into account the TCI, the specific methanol production cost and the cost-revenues balance. In addition, the influence on the economic results of oxygen selling is investigated for each solution.

2.1. Economic assumptions

To perform the analysis, the German economic scenario is chosen and the following economic assumptions are considered:

- **Electrical energy cost** represents a term of primary importance to find out the optimal system configuration. In this analysis, the electrical energy to feed electrolyzers is assumed to be purchased from the grid at 30 €/MWh, the average market price in Germany for 2015 [23,26,27]
- **Methanol selling price** is another parameter that depends on the economic scenario where the plant is going to operate; in this analysis, the European methanol market is chosen as target. The average methanol market value in Europe between 2014 and 2015 was about 400 €/ton, but as the methanol market is going to expand its market value may become higher as reported in [24,25]. Thus different market values (from a minimum value of 400 €/ton to a maximum value of 800 €/ton) are considered in the analysis in order to take into account possible future scenarios.
- **Oxygen selling price** highly depends on its diffusion in the market, which is related to local conditions, applications, etc. In the case under analysis, Oxygen price is assumed 100 €/ton, which represents the minimum selling price for industrial use of oxygen [21] (rates are higher for medical use). It is worth noting that the purity of Oxygen produced by electrolyzers (> 99.9%) is sufficient for industrial applications, therefore no further purification treatments are needed.
- **Electrolysers capital cost** highly depends on the technology developments: at the present moment, market costs for large size PEM electrolyzers (15 MW) is estimated 1000 €/kW, while for medium size PEM electrolyzers (1 MW) is 1500 €/kW [22]. Cost function can be represented as follows:

$$C = 1.5 \cdot 10^6 \cdot (P)^{0.85} \quad (4)$$

where P represents the installed power. However, with their wide diffusion in the market, costs can be reduced significantly; future scenarios with a possible reduction in terms of capital costs of 10% 20% and 30%

respectively are investigated.

- **Methanol reactor capital cost** is estimated by literature values, referred to the commercial large scale plants. In particular, the cost function reported below is based on the plant capacity of 54 ton/h (in terms of gas entering the reactor) [17]. A standard factor of 0.65 is employed in order to consider the size effect on investment costs [24].

$$C = 14.2 \cdot 10^6 \cdot \left(\frac{M_{in}}{54000} \right)^{0.65} \quad (5)$$

- **Carbon Capture and Storage (CCS) system capital cost** is estimated by literature values, referred to commercial large scale plants. In particular, the cost function reported below is based on the plant capacity of 2808 ton/h (in terms of gas entering the separator) [28]. A standard factor of 0.65 is employed in order to consider the size effect on investment costs [20].

$$C = 75.45 \cdot 10^6 \cdot \left(\frac{M_{in}}{2.808 \cdot 10^6} \right)^{0.65} \quad (6)$$

- **Plant lifetime** is assumed equal to 15 years to be conservative, considering the lifetime of the electrolyzers [22], which represent the most expensive plant components.
- **Methanol plant equivalent operative hours** are assumed equal to 8000 h per year, which represents a typical value for this kind of plants. In fact, due to the high inertia of chemical reactors, the methanol synthesis plant should operate at nominal conditions for the whole year, if possible.
- **Inflation** is assumed equal to 0.4%, which represents the average value in Germany in 2015, as reported in [29].
- **Average income tax rate** is assumed equal to 19%, which represents the average value in Germany in 2015, as reported in [29].

Table 3 reports the main data assumed for the present thermo-economic analysis. All these data values are inputs for the W-ECOMP software.

3. Thermo-economic results

In the present analysis, an average electrical energy cost is

Table 2
Main thermo-dynamic assumption: mass balance and energy balance.

	Mass Balance	Energy Balance
Electrolyser		
Power installed		1 MW
Energy consumption (+ auxiliary)		1.03 MWh
Hydrogen outlet	19 kg/h	
Oxygen outlet	152 kg/h	
Water consumption	285 kg/h	
CCS system		
Flue gas in	700 kg/h	
wt% CO ₂ in flue gas	20%	
CO ₂ out	140 kg/h	
Thermal energy consumption		117 kWh _{th}
Electrical energy consumption		7 kWh
Methanol reactor		
Mixture inlet	159 kg/h	
Methanol outlet	97 kg/h	
Thermal energy outlet (based on the heat of reaction)		31 kWh _{th}
Compressors		
CO ₂ compression (from 2 up to 30 bar)	140 kg/h	9.8 kWh
H ₂ + CO ₂ compression (from 30 up to 80 bar)	159 kg/h	13 kWh

Table 3
Economic data.

Economic scenario parameters	
Economic data reference year	2015
Construction starting year	2015
Construction time	1 year
Plant lifetime	15 years
Depreciation time	10 years
Inflation rate	0.40%
Nominal escalation rates	2.50%
O & M factor	1.04%
Average income tax rate	19%
Financing fraction	
Debts	50%
Preferred stocks	35%
Common equities	15%
Annual rate	
Debts	5%
Preferred stocks	5%
Common equities	5%
Discount rate	0.40%

Table 4
Mass flows for the different plant sizes.

		4000 t/yr	10,000 t/yr	50,000 t/yr
	Methanol capacity plant			
ELECTROLYSER	Power	5 MW	13 MW	63 MW
	H ₂ out	94 kg/h	247 kg/h	1197 kg/h
	O ₂ out	752 kg/h	1976 kg/h	9576 kg/h
CCS SYSTEM	Flue gas in	3440 kg/h	9050 kg/h	43890 kg/h
	wt% CO ₂ in flue gas	20 wt%	20 wt%	20 wt%
	CO ₂ out	688 kg/h	1810 kg/h	8778 kg/h
METHANOL REACTOR	Mixture in	782 kg/h	2058 kg/h	9975 kg/h
	H ₂ :CO ₂	1:3	1:3	1:3
	conv efficiency	0.96	0.96	0.96
	Methanol out	483 kg/h	1265 kg/h	6130 kg/h

considered and the influence of methanol selling price is investigated in order to determine which options can be feasible from the economic point of view. The influence of oxygen selling is evaluated as well. The results are compared in terms of DPBP, since it represents one of the most important parameters for thermo-economic evaluations. Assuming a plant lifetime of 15 years, all the solutions characterized by a DPBP higher than 10 years are considered not acceptable. Table 4 summarizes mass flows and installed power (in terms of electrolyzers) for the

different plant sizes.

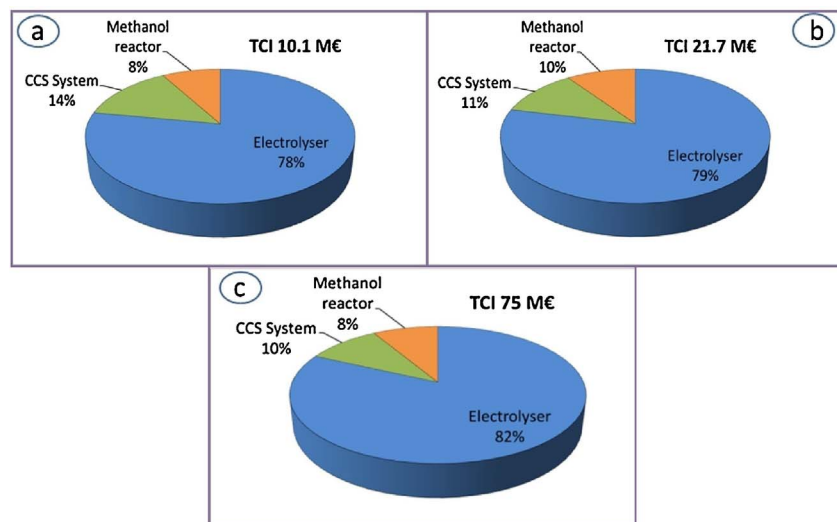
Fig. 3 reports the TCI distribution among the different plant components: electrolyzers represent the most significant voice, even if the capital costs of methanol reactor and CO₂ sequestration system have an impact as well. Since the coal fired power plant was already installed, its capital cost is not taken into account.

Three different system sizes are analyzed, it is worth observing the influence of size on TCI: the capital cost for the small size system (4000 ton/year of CH₃OH) is about 10 M€, corresponding to 2500 € per ton/year of CH₃OH; however, specific capital costs are lower for the medium size of 10,000 ton/year of CH₃OH (2170 € per ton/year of CH₃OH) and for the large size of 50,000 ton/year of CH₃OH (1500 € per ton/year of CH₃OH).

Fig. 4 reports the annual costs and revenues contribution for the different plant sizes. This first analysis is performed considering the average methanol price in the European market in 2014 and 2015, equal to 400 €/ton [11]. Revenues are generated by methanol and oxygen selling, while costs are mainly due to annual fixed costs (plant depreciation) and variable costs are mainly related to electrical energy to feed the power-to-fuel plant. The difference between revenues and costs is the annual cash flow: it is worth noting that, in case of low and medium sizes cash flows are quite low, therefore these solutions appear not fully convenient. On the other hand, large size plant seems the most promising (cash flow is about 6 M€/year), even if the investment is the highest as well (about 75 M€, see Fig. 3).

The series of analysis reported in Fig. 5 is performed in consideration of not selling the large amount of Oxygen produced by the electrolyzers and thus venting it to the atmosphere. The simulations are carried out for the three sizes in order to determine the minimum methanol selling price to achieve economically feasible results (DPBP lower than 10 years). Considering methanol market prices of 400 and 500 €/ton, no solutions are available; at 600 €/ton, only the large scale plant would be feasible; the only attractive solutions would be the ones with a high methanol selling price (at least 700 €/ton), which is not realistic in the present market scenario. For this reason, not selling oxygen would not be a feasible solution from the economic point of view, at least in the current market scenario.

Considering Oxygen selling at the conservative price of 100 €/ton, the results are significantly improved as shown in Fig. 6. For electricity cost of 0.03 €/kWh at the current methanol price of 400 €/ton, the plant would be still not economically feasible, but considering methanol selling price at 500 €/ton the large scale plant would be an acceptable solution (DPBP equal to 10 years); for larger selling price of 600 €/ton even the lower size plant configurations are economically feasible.

**Fig. 3.** Capital costs distribution for 4000 (a) 10,000 (b) and 50,000 (c) ton/year CH₃OH capacity plants.

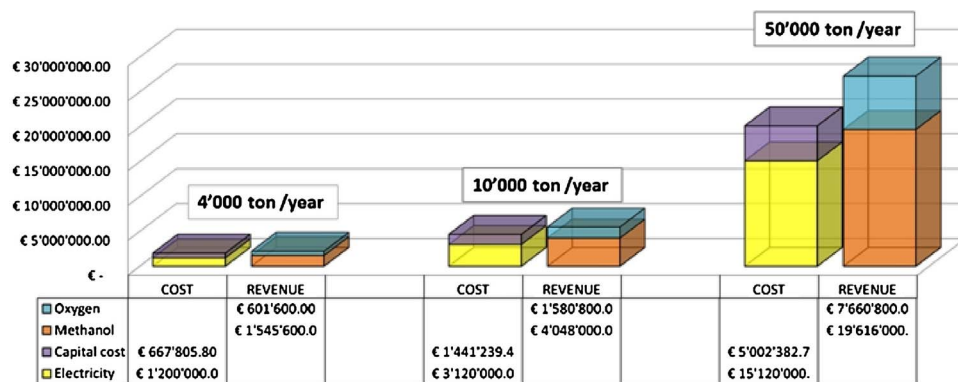


Fig. 4. Annual costs and revenues for 4000 10,000 and 50,000 ton/year CH₃OH capacity plants.

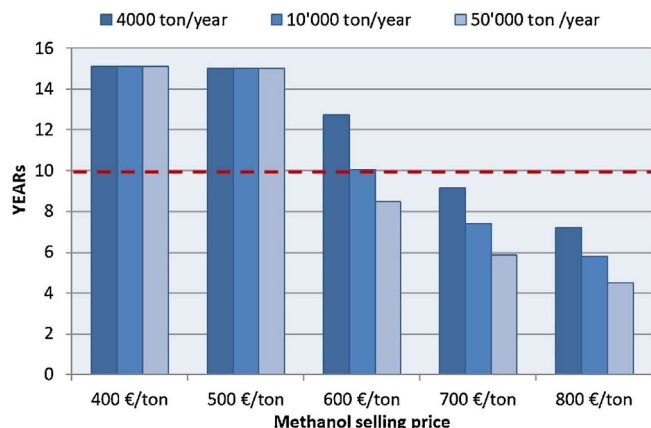


Fig. 5. Economic results (in Case of O₂ vent to atmosphere).

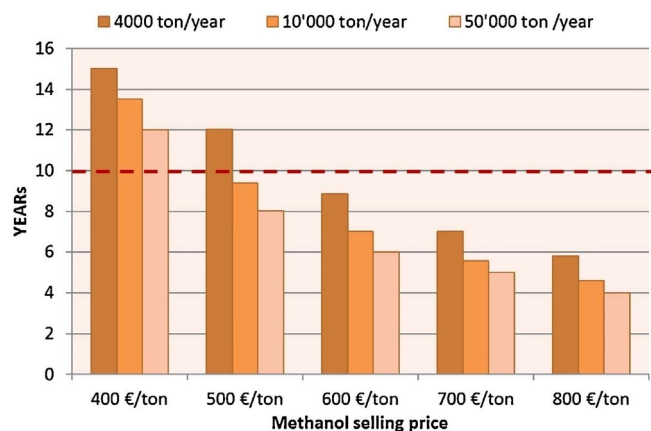


Fig. 6. Economic results (in case of O₂ selling at 100 €/ton).

As development in terms of technology and further diffusion of these kinds of plants in the market is going to increase, capital costs related to electrolyzers and other plant components are going to get reduced. Fig. 7 shows the trend of predictable reduction (different rates of reduction) in the next years. At last, it is interesting to note that, in particular for large size plants, economic results would be acceptable with a DPBP lower than 10 years even considering the current methanol selling price of 400 €/ton. In a future scenario with higher methanol demand and consequently higher market price (450 or 500 €/ton), even the lower size solutions would be feasible. In a foreseeable favourable scenario (decreasing of capital costs and higher methanol selling prices), DPBP lower than 5 years would be possible.

It is worth mentioning that this analysis has been performed considering a constant cost for electrical energy, as the aim was to understand the main influence of the different parameters considered

above on the economic results. The most important parameters can be summarized as:

- Selling the large amounts of oxygen produced in the electrolyzers, in order to increase the economic performance;
- The future economic scenario in terms of methanol selling price (affected by market demand);
- The future economic scenario in terms of capital costs (affected by development in terms of technology and market diffusion);
- Size influence: larger size plants lead to higher capital investment but lower DPBP.

4. Conclusions

In this paper, the economic feasibility of plants for methanol synthesis from hydrogen (produced by pressurized water electrolyzers) and carbon dioxide (sequestered by fossil fuel power plants), has been investigated. A preliminary thermo-economic analysis has been performed in the framework of the H2020 EU project MefCO₂. The German economic scenario has been investigated, taking into proper account the difference of several energy and market parameters in order to evaluate the cases in which the methanol plant is economically feasible. The analysis has been performed considering real market costs and cost functions for the different plant components by using a dedicated software for the thermo-economic analysis. The analysis was performed considering three different plant sizes (4000 10,000 and 50000 ton/year of CH₃OH produced) and investigated the influence of different parameters, in particular:

- Methanol selling price: the market of methanol is going to expand and the price is strongly variable;
- Oxygen selling option for the industrial applications: in the process of water electrolysis it is produced in large amounts and with high purity (99.9%);
- Capital cost of PEM electrolyzers: expected technology developments in the next years may allow for a significant reduction in terms of capital costs.

From the obtained results, the following conclusions can be drawn:

- Considering the high capital cost of the plant, Oxygen selling is mandatory in order to obtain economically feasible results, all the solutions with oxygen vent to the atmosphere are proved to be not feasible;
- PEM electrolyzers are the most expensive components of the plant, affecting for more than 75% on the TCI: at their current market cost the plant would be feasible only in case of CH₃OH selling at more than 600 €/ton;
- Considering a reduction of TCI due to expected developments in PEM electrolyzers technology, different solutions would be

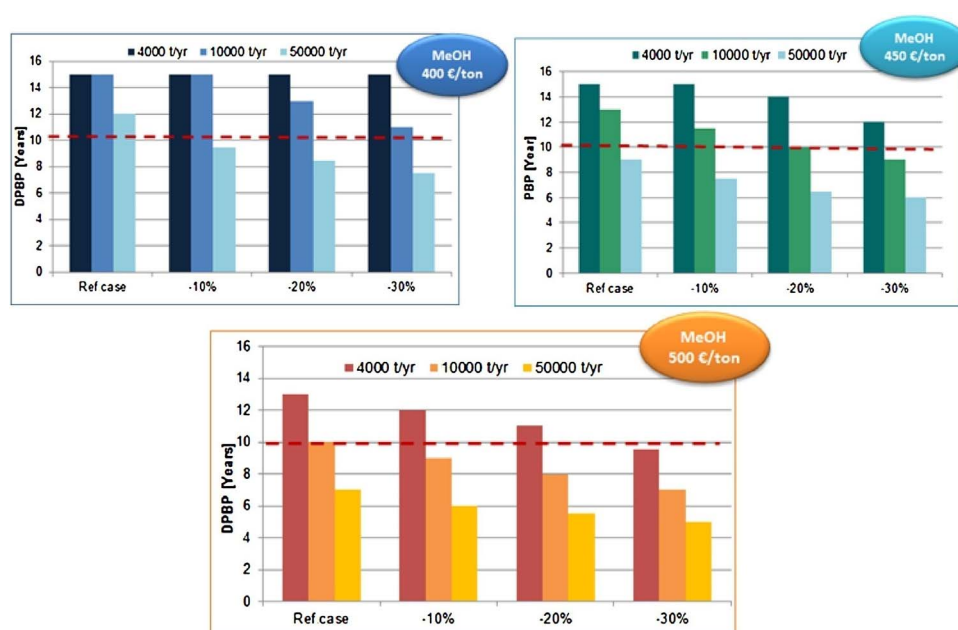


Fig. 7. Influence of capital costs reduction on results for different CH₃OH selling prices.

economically feasible (see Fig. 6);

- Larger plant sizes (50,000 ton/year of CH₃OH) lead to higher capital investment but allow for lower DPBP therefore, the attention will be focused on these plants for the future analysis.

It is worth observing that in case of the large size solution, a considerable amount of CO₂ (71,600 ton/year) is sequestered by fossil plants and used for methanol production, which in turn allow for a reduction in operating plant costs in terms of CO₂ emission taxation. Moreover, comparing this solution with the traditional production by natural gas steam reforming process, a considerable fuel saving of about 30000 ton/year [30] can be obtained: this value corresponds to the annual consumption of about 12 domestic users (assuming an average NG consumption of 2000Sm³/year per user [30]).

The results obtained from this first analysis show a significant potential for innovative low-carbon footprint methanol production, suggesting the importance of O₂ selling and investment in large size plants.

Finally, another important advantage of these plants that will be the object of research in the next step of the EU project, is the possibility to employ these systems to reduce CO₂ emissions and, at the same time, to offer an ancillary service for national grid stabilization by providing the energy storage (in chemical form), which is able to absorb the production peaks due to unpredictable RES. In this way, these plants can present a potential opportunity to solve a critical aspect of fluctuations in grid power, due to the increased installed capacity of unpredictable RES (in particular solar and wind) in most of the EU Countries (i.e. Germany, Italy and Spain).

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